



MTEN 6020

Project Check-In V

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I. Building the Fe-Ni-Nb Ternary .tdb

Overview



Ternary System Selected: **Fe-Ni-Nb**

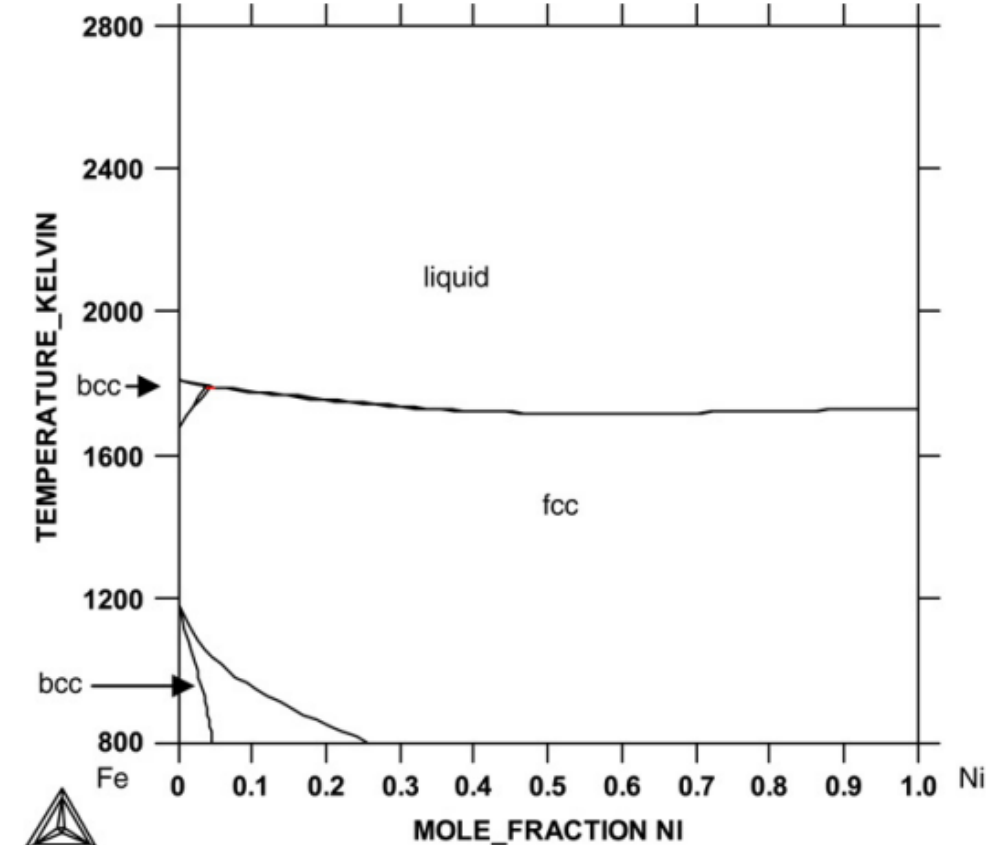
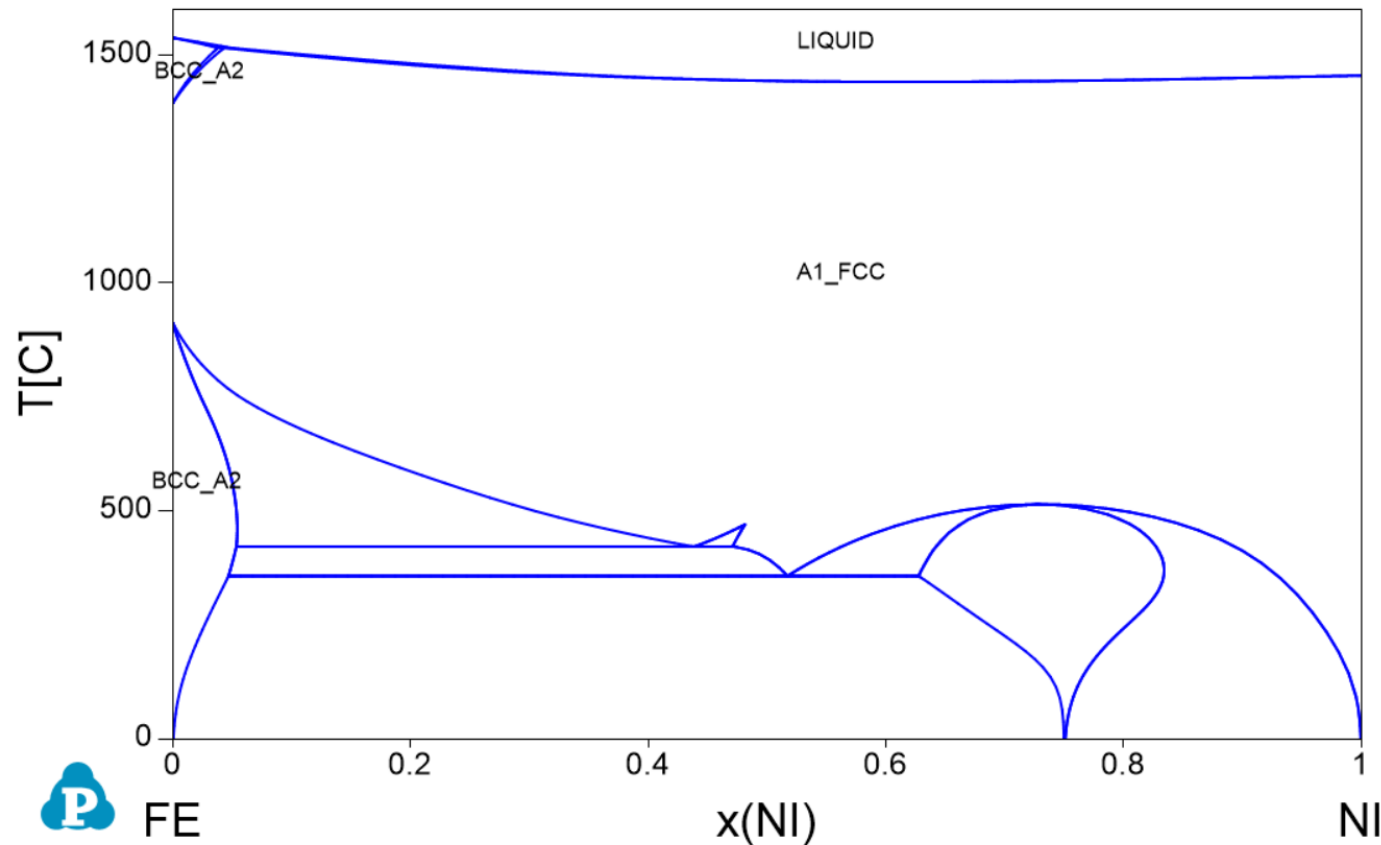
Rationale:

- Relevance to IN718 primary phase transformation ($\gamma \rightarrow \gamma''$)
- Binary thermodynamic databases are well-established
- Ternary thermodynamic database has been evaluated as well
- Evaluated mobility data for γ and γ'' phases

2.1.1 Checklist

- ☐ Collect and review binary thermodynamic databases for A-B, A-C, and B-C
- ☐ Verify each of the starting binary phase diagrams reproduce: Phase stability ranges, invariant reactions (eutectic, peritectic, etc.), reasonable liquidus/solidus shapes
- ☐ Create a preliminary ternary database from your three binary databases
- ☐ Demonstrate that the ternary database is functional by plotting ternary sections at three temperatures and verifying that the binaries in your ternary database match the binary databases

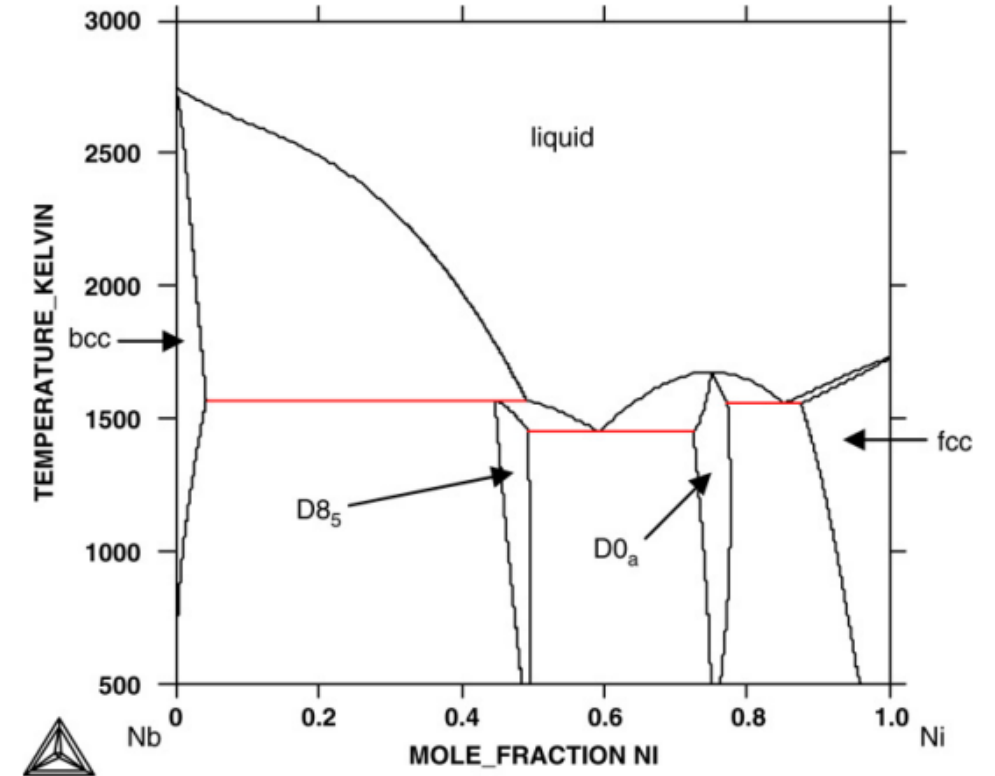
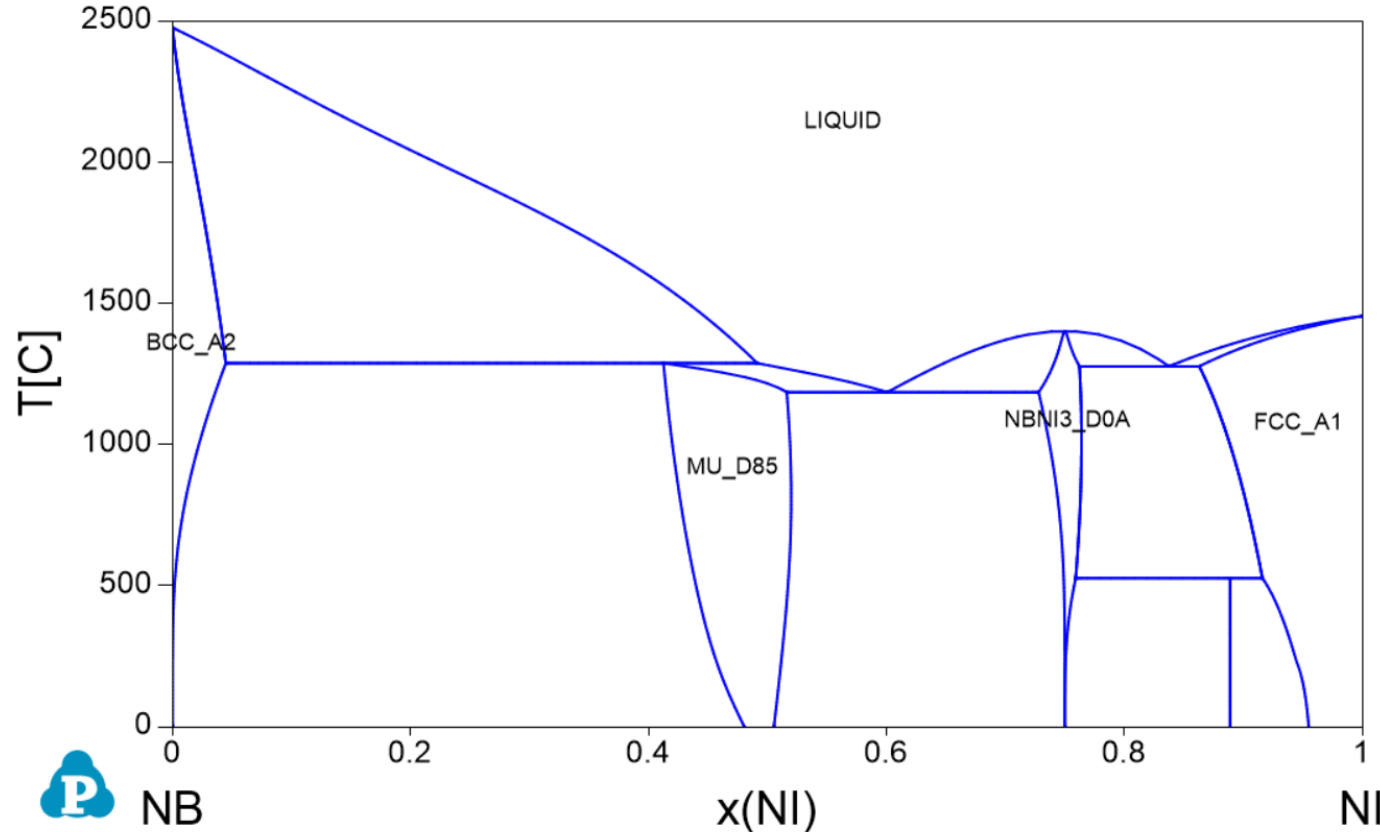
Fe-Ni Binary Phase Diagram (Binary .tdb)



TDB Referenced: Z.-T. Zhang, J.-J. Wang, W. Liu, X.-G. Lu, J. Iron Steel Res. Int., 30, 660-76(2023).

<http://dx.doi.org/10.1016/j.calphad.2008.10.005>

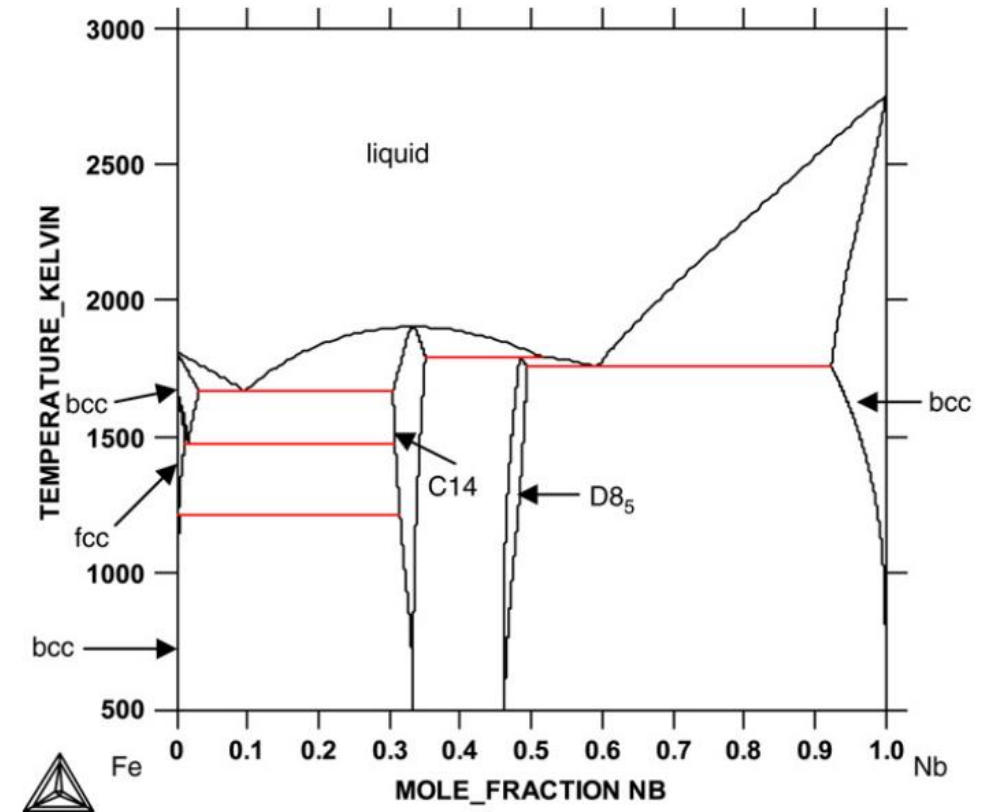
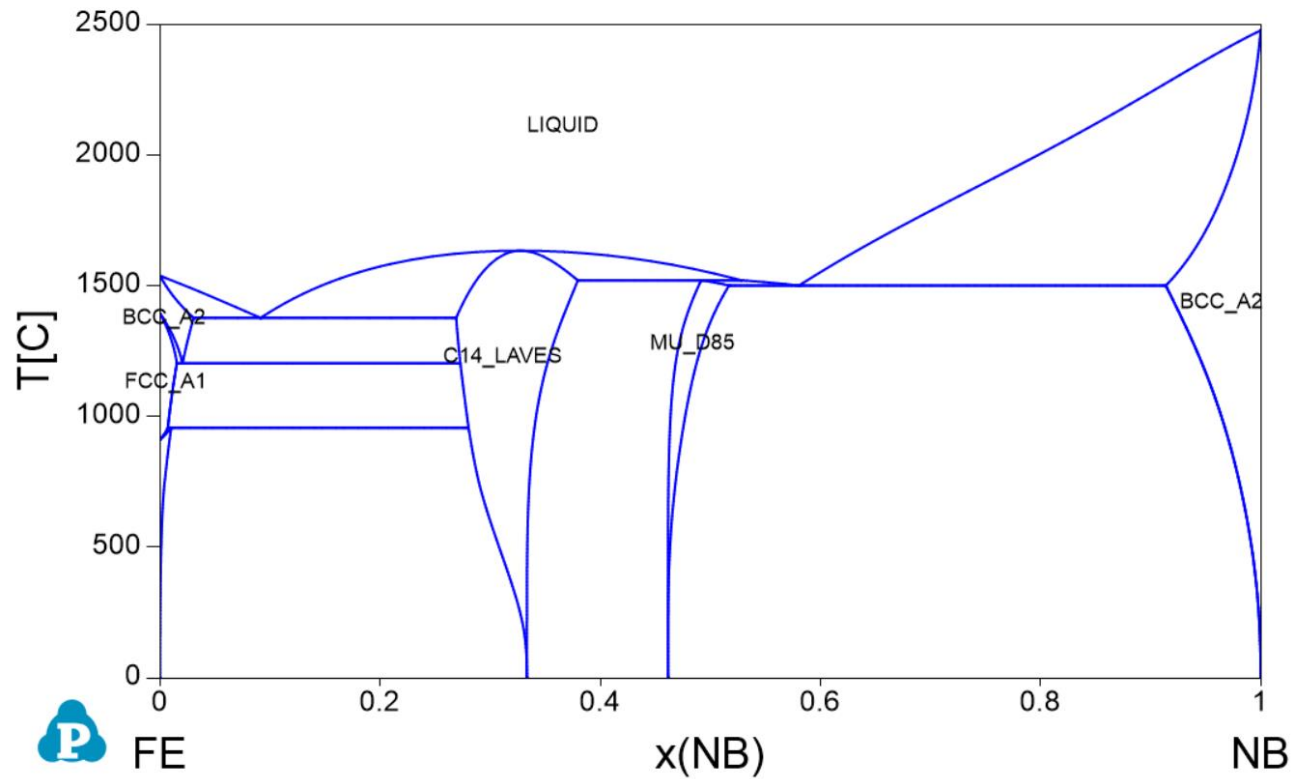
Ni-Nb Binary Phase Diagram (Binary .tdb)



TDB Referenced: H. Sun, S.-L. Shang, R. Gong, B.J. Bocklund, A.M. Beese, Z.-K. Liu, Calphad, 82, 102563(2023).

<http://dx.doi.org/10.1016/j.calphad.2008.10.005>

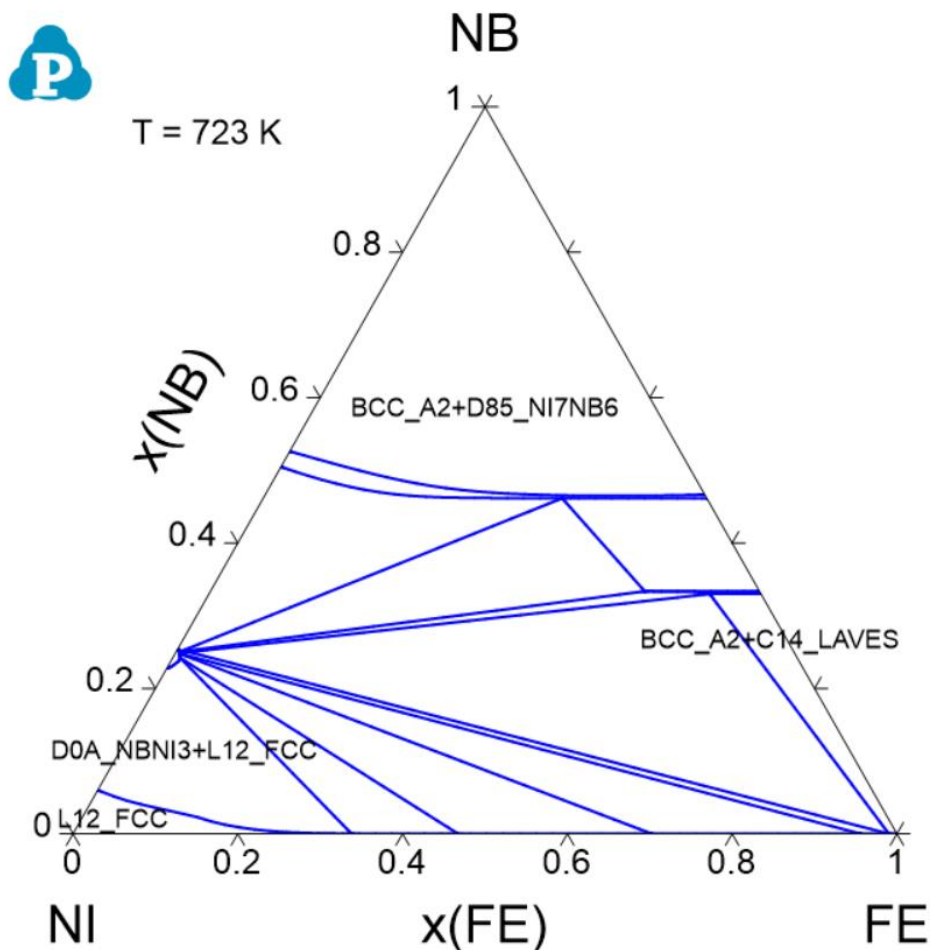
Fe-Nb Binary Phase Diagram (Binary .tdb)



TDB Referenced: H. Sun, C. Wang, S.-L. Shang, A.M. Beese, J.-C. Zhao, Z.-K. Liu, Acta Mater., 268, 119747(2024).

<http://dx.doi.org/10.1016/j.calphad.2008.10.005>

Fe-Ni-Nb Ternary Phase Diagram (T = 723 K)



TDB Referenced: Fe-Nb-Ni_Mathon_v3.tdb

(a database I recreated using Appendix B of →)

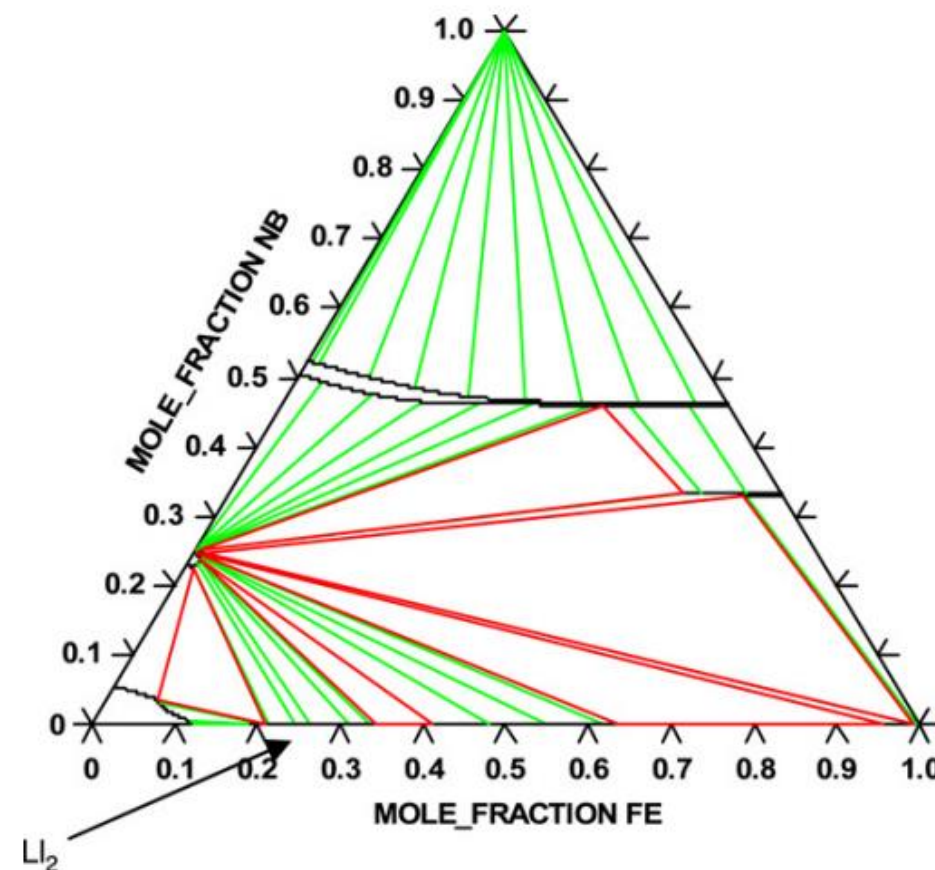
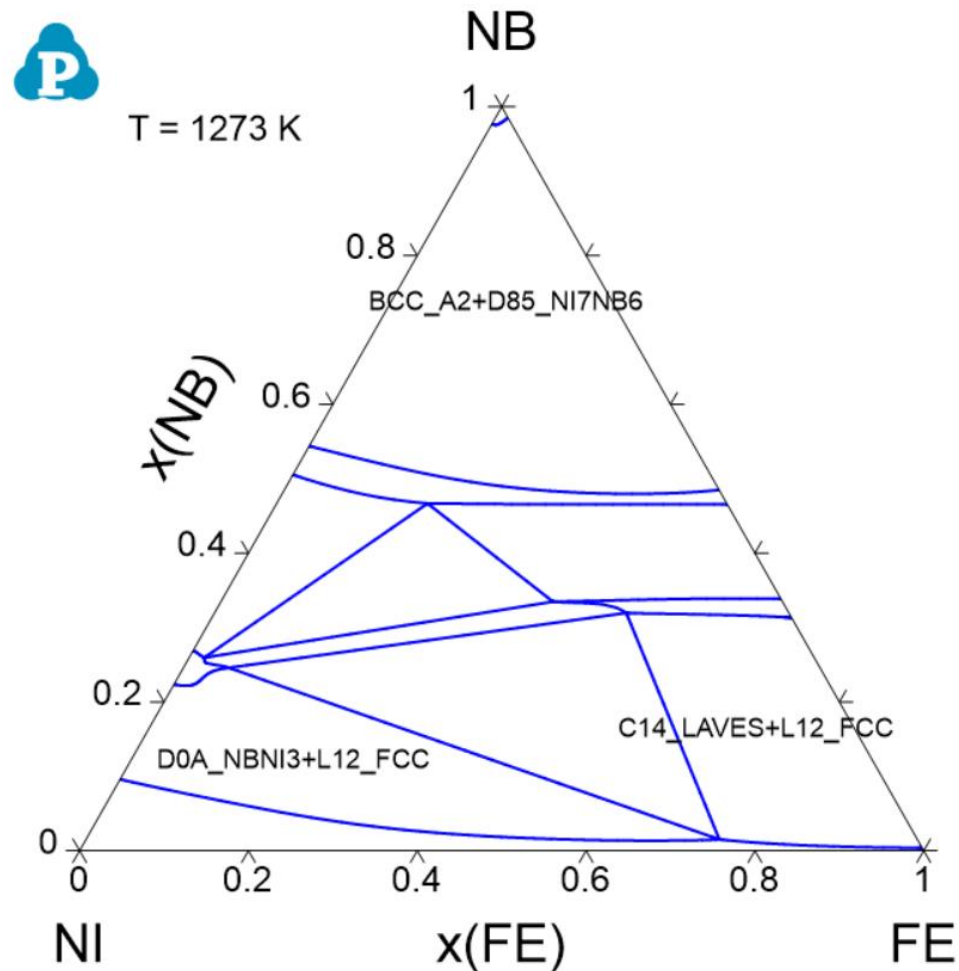


Fig. 14. Isothermal section at 723 K of the ternary Fe–Nb–Ni phase diagram.

<http://dx.doi.org/10.1016/j.calphad.2008.10.005>

Fe-Ni-Nb Ternary Phase Diagram (T = 1273 K)



TDB Referenced: Fe-Nb-Ni_Mathon_v3.tdb

(a database I recreated using Appendix B of →)

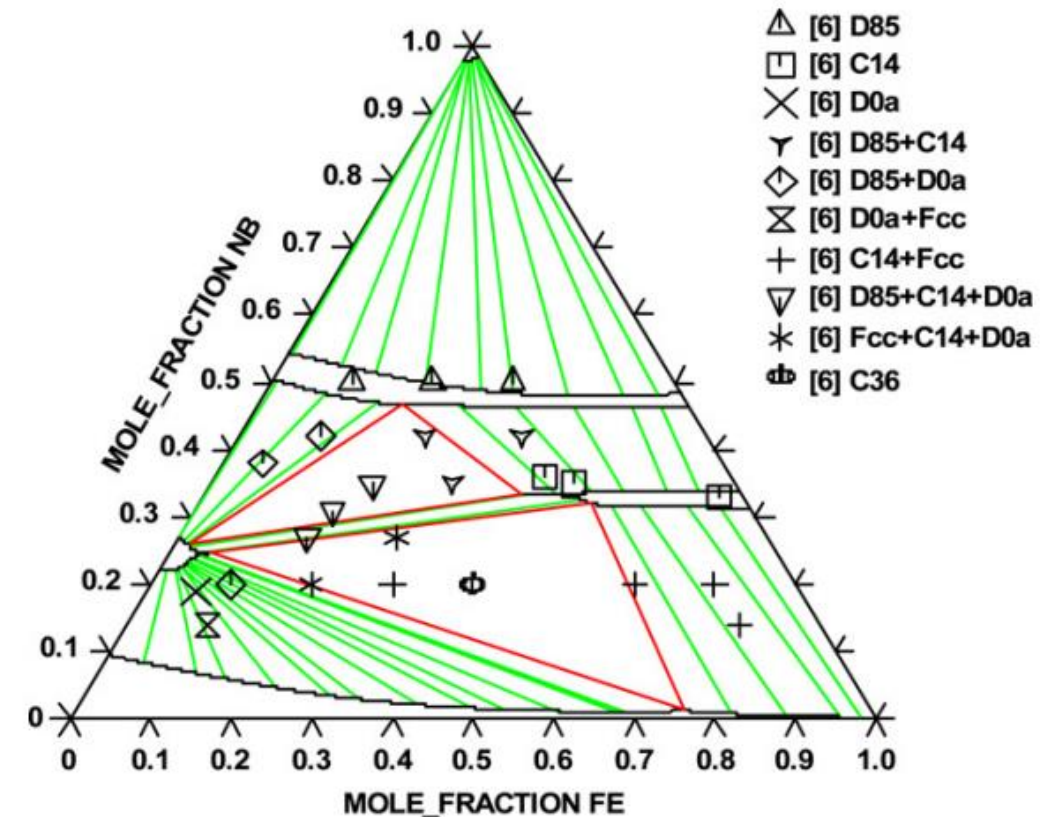
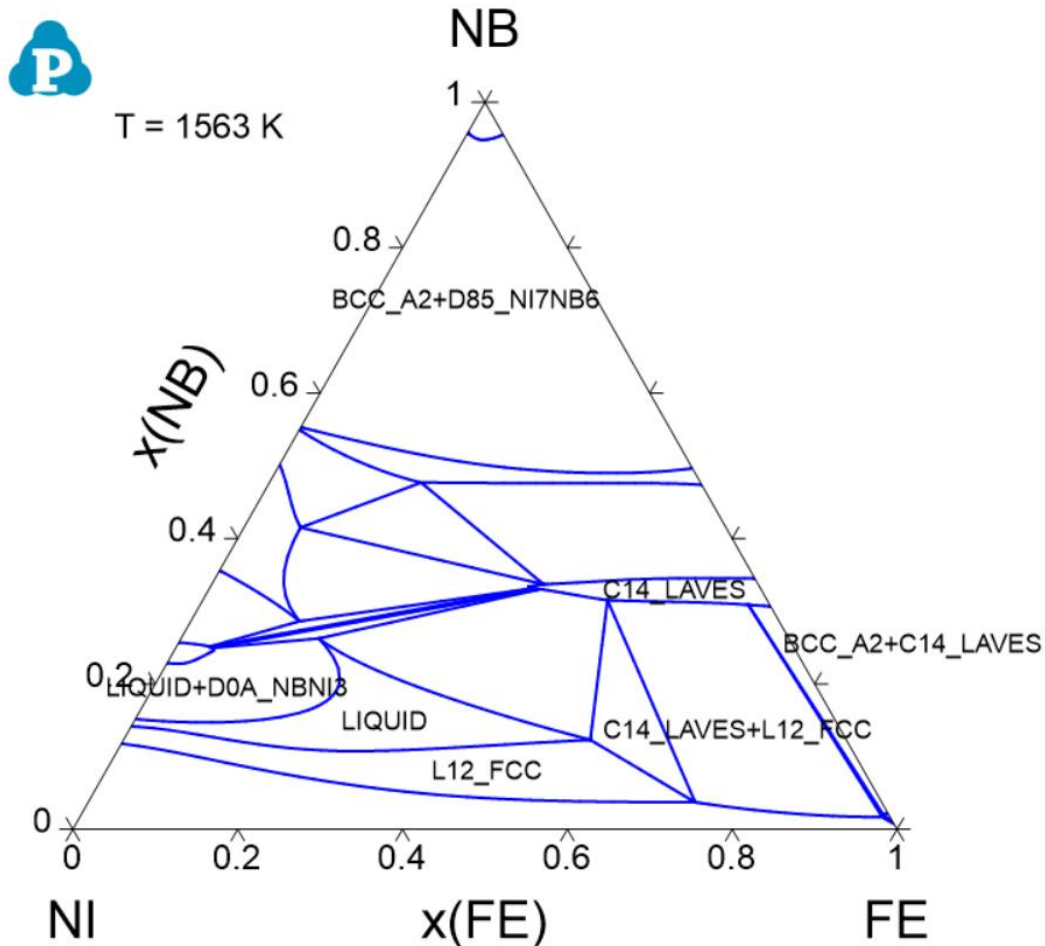


Fig. 11. Isothermal section at 1273 K of the Fe-Nb-Ni ternary phase diagram.

<http://dx.doi.org/10.1016/j.calphad.2008.10.005>

Fe-Ni-Nb Ternary Phase Diagram (T = 1563 K)



TDB Referenced: Fe-Nb-Ni_Mathon_v3.tdb

(a database I recreated using Appendix B of →)

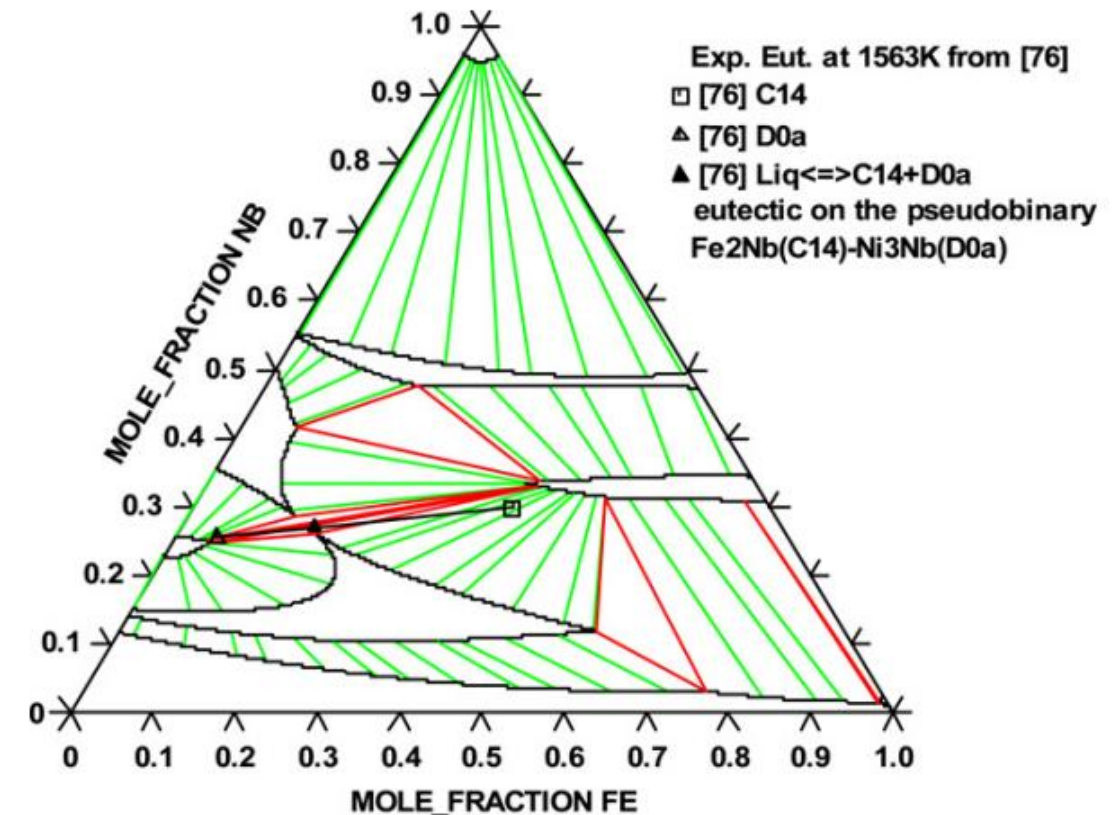
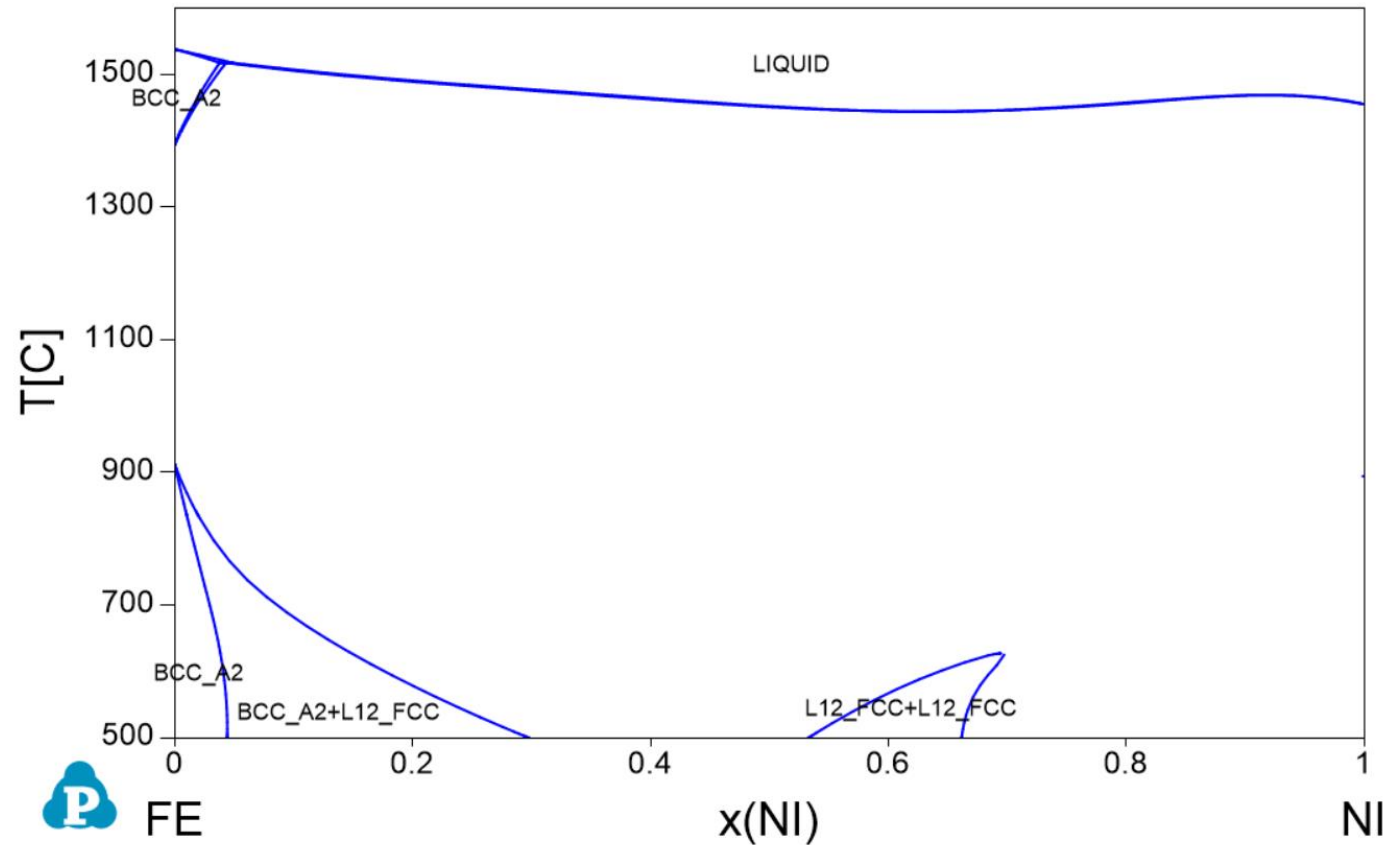


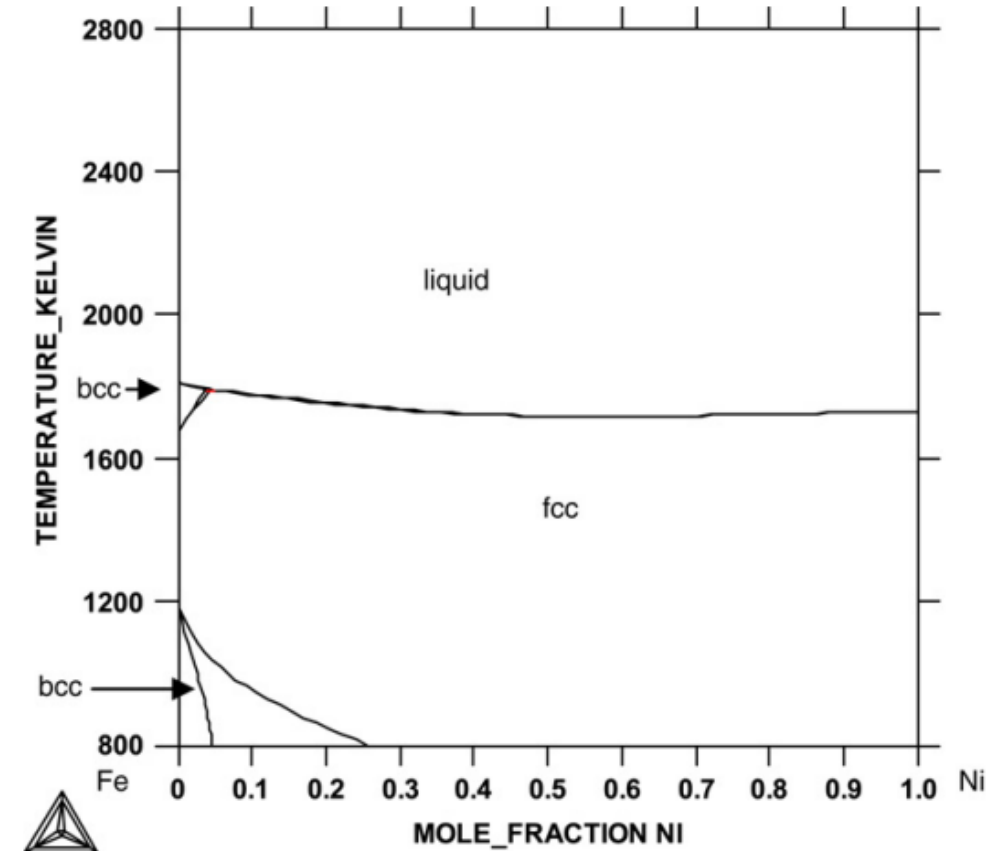
Fig. 8. Isothermal section at 1563 K of the Fe-Nb-Ni ternary phase diagram.

<http://dx.doi.org/10.1016/j.calphad.2008.10.005>

Fe-Ni Binary Phase Diagram (Ternary .tdb)

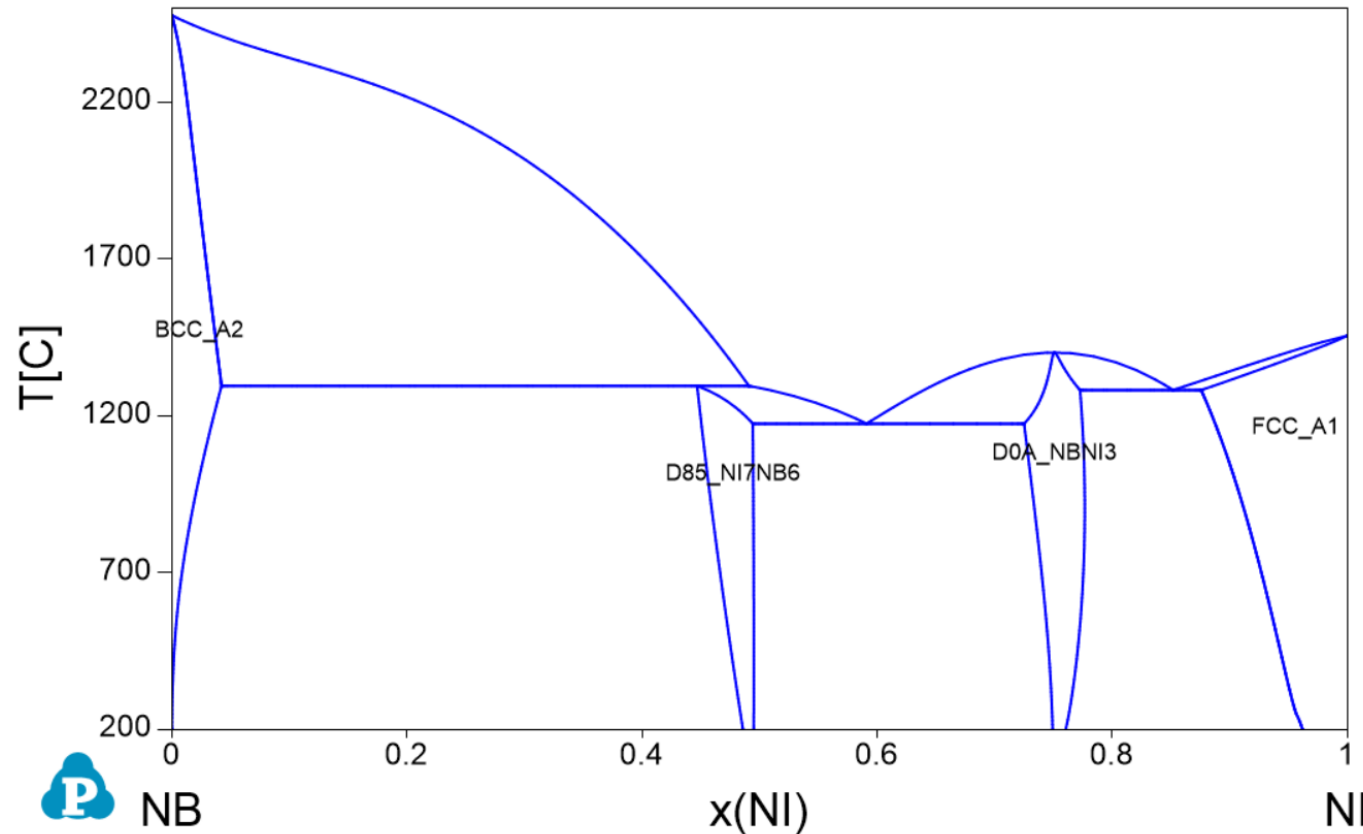


TDB Referenced: Fe-Nb-Ni_Mathon_v3.tdb

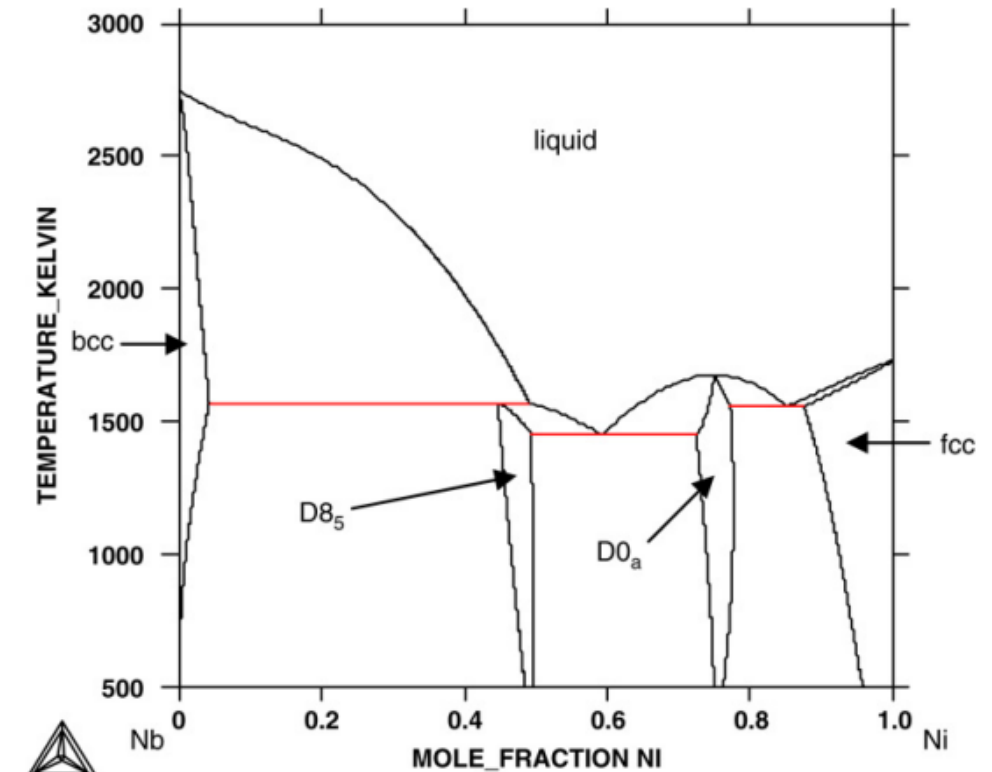


<http://dx.doi.org/10.1016/j.calphad.2008.10.005>

Ni-Nb Binary Phase Diagram (Ternary .tdb)

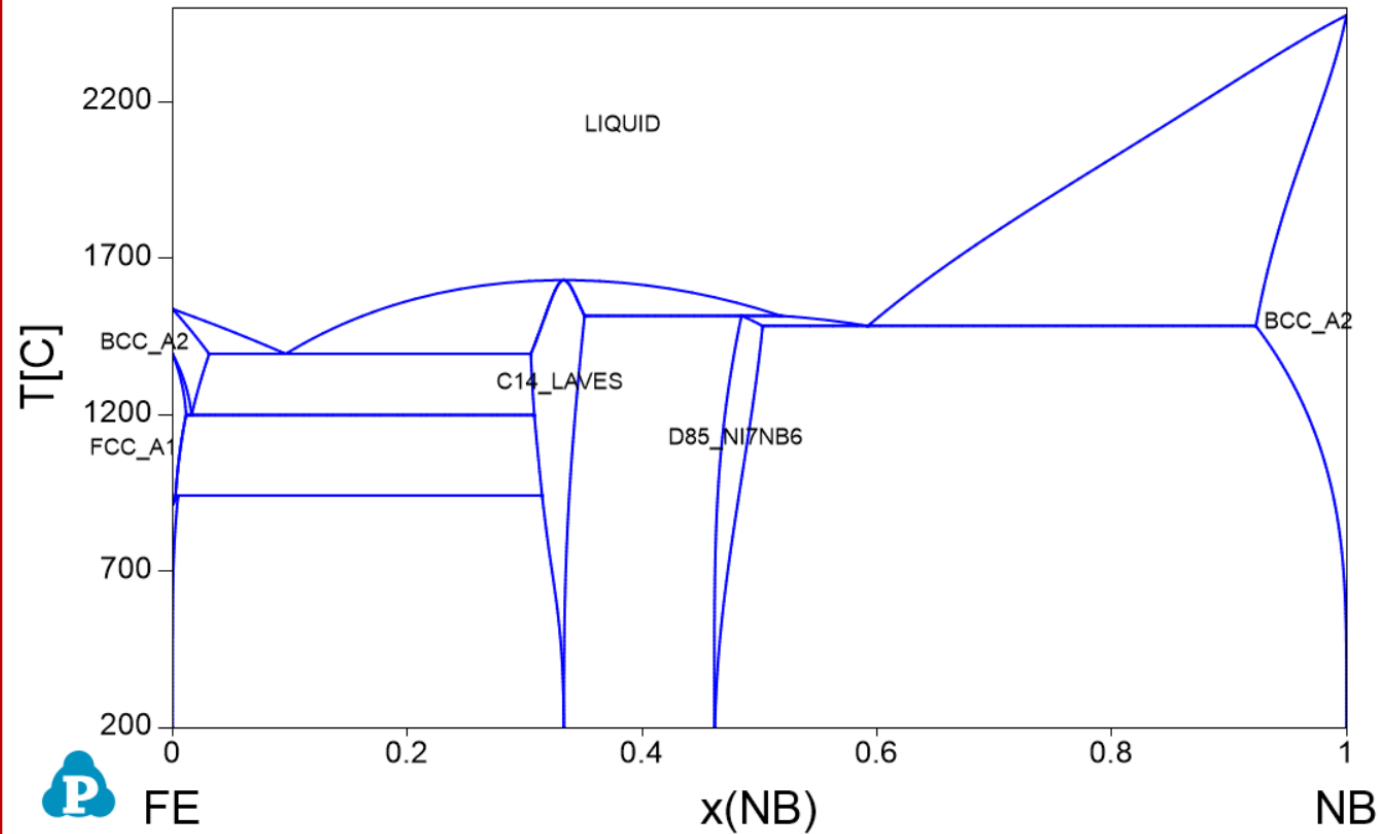


TDB Referenced: Fe-Nb-Ni_Mathon_v3.tdb

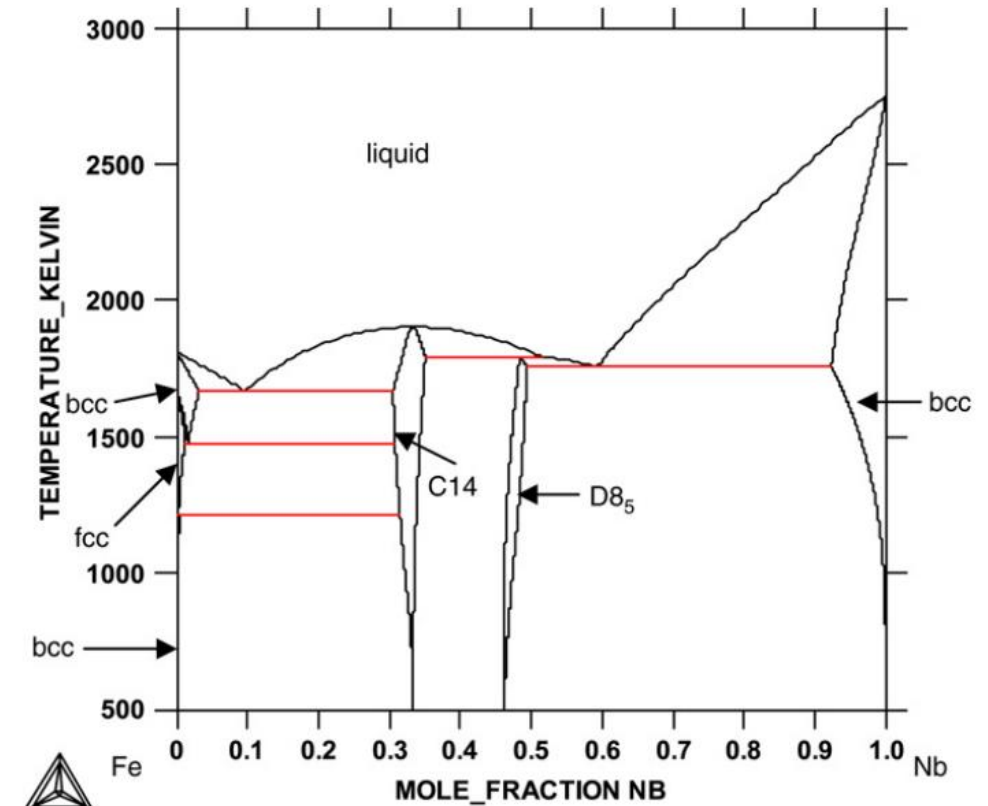


<http://dx.doi.org/10.1016/j.calphad.2008.10.005>

Fe-Nb Binary Phase Diagram (Ternary .tdb)



TDB Referenced: Fe-Nb-Ni_Mathon_v3.tdb



<http://dx.doi.org/10.1016/j.calphad.2008.10.005>

Key Outcomes



- Mathon et al. Fe-Ni-Nb database was formatted for Thermo-Calc and could not be imported directly into Pandat 2025
- Successfully translated the Mathon et al. database so it can be used with Pandat 2025 and reproduced the established binary and ternary phase diagrams using the ternary database

II. Finding Mobility Data

Overview



2.2.1 Checklist

- ☐ Identify each phase in your tdb for which you will need to define mobility parameters
- ☐ Gather diffusion data for each binary: Tracer, interdiffusion, etc for each phase
- ☐ Gather any diffusion data for ternary that may be available

- The Fe-Ni-Nb ternary system was selected as a reduced-order, ternary approximation of IN718
- The precipitates we would like to capture in the eventual TTT and CCT diagrams calculated for this project include the γ'' and δ phase
- γ'' and δ are different crystal structures of Ni_3Nb that precipitate out of the γ matrix
- Therefore, the mobility data that has the most relevance are the diffusivities of Fe, Ni, and Nb in γ -Ni

Mobility Data Search Strategy



Mobility Database Governing Equation:

$$\underbrace{RT \ln(M_i \cdot \Gamma_i)}_{\text{Diffusivity}} = \underbrace{\sum_p x_p \phi_i^p}_{\text{End-Member Parameters}} + \underbrace{\sum_p \sum_{q > p} x_p x_q \left[\sum_r \phi_i^{p,q} (x_p - x_q)^r \right]}_{\text{Interaction Parameters}}$$

- We will assume the interaction parameters, $\phi_i^{p,q}$, are zero (i.e. a dilute solution model)
 - Reduces how much experimental data is required to build a mobility database
- Tracer diffusivities of species i in component p required for the phases of interest (γ -Ni)

Mobility Data Search Strategy



To build a mobility database that can capture the phase transformations of interest ($\gamma \rightarrow \gamma'' \rightarrow \delta$), the following mobility data is required:

End-Members for γ Phase				
Species i	$\phi_i^{\text{Ni, FCC}}$	$\phi_i^{\text{Fe, FCC}}$	$\phi_i^{\text{Nb, FCC}}$	
	Ni	$D_{0\text{Ni,FCC}}^i$	$D_{0\text{Fe,FCC}}^i$	$D_{0\text{Nb,FCC}}^i$
	Fe	$Q_{\text{Ni,FCC}}^i$	$Q_{\text{Fe,FCC}}^i$	$Q_{\text{Nb,FCC}}^i$
	Nb			

Mobility Data Found for γ -Ni Matrix



End-Members for γ Phase			
Species i	$\phi_i^{\text{Ni,FCC}}$	$\phi_i^{\text{Fe,FCC}}$	$\phi_i^{\text{Nb,FCC}}$
	Ni $D_0 = 3.44 \text{ cm}^2 \text{ s}^{-1}$ $Q = 290.0 \text{ kJ mol}^{-1}$ [5]	$D_0 = 1.12 \text{ cm}^2 \text{ s}^{-1}$ $Q = 283.5 \text{ kJ mol}^{-1}$ [5]	$D_0 = 3.44 \text{ cm}^2 \text{ s}^{-1}$ $Q = 290.0 \text{ kJ mol}^{-1}$ [5]
	Fe $D_0 = 9.20 \text{ cm}^2 \text{ s}^{-1}$ $Q = 297.7 \text{ kJ mol}^{-1}$ [5]	$D_0 = 0.72 \text{ cm}^2 \text{ s}^{-1}$ $Q = 278.7 \text{ kJ mol}^{-1}$ [5]	$D_0 = 9.20 \text{ cm}^2 \text{ s}^{-1}$ $Q = 297.7 \text{ kJ mol}^{-1}$ [5]
	Nb $D_0 = 0.0104 \text{ cm}^2 \text{ s}^{-1}$ $Q = 202.59 \text{ kJ mol}^{-1}$ [6]	$D_0 = 0.75 \text{ cm}^2 \text{ s}^{-1}$ $Q = 264 \text{ kJ mol}^{-1}$ [7]	$D_0 = 0.0104 \text{ cm}^2 \text{ s}^{-1}$ $Q = 202.59 \text{ kJ mol}^{-1}$ [6]

* The end-member parameter for the Nb component in the γ matrix is assumed to be equal to the end-member parameter of the Ni component. This assumption is necessary because Nb is BCC for all temperatures up to its melting temperature, so tracer diffusivity in FCC Nb cannot be evaluated experimentally.

III. Calculating Binary & Ternary Diffusion Profiles

Overview



Key Outcomes:

- Integrate mobility data found for check-in 2 to the Fe-Ni-Nb .tdb from check-in 1
- Verified mobility data implemented correctly, and physically reasonable, by running simulations of binary and ternary diffusion simulations
- Simulations targeted the compositions relevant to the $\gamma \rightarrow \gamma'' \rightarrow \delta$ phase transformation that is intending to be captured by the final TTT and CCT diagrams

2.3.1 Checklist

- ☐ Implement ternary mobility expressions for each phase
- ☐ Decide whether ternary mobility interaction terms are required
- ☐ Verify Onsager matrices are positive definite and symmetric
- ☐ Run ternary diffusion simulations to show (1) composition profile evolution and (2) diffusion paths in ternary space
- ☐ Check for expected features in diffusion profiles (e.g., uphill diffusion, partitioning at phase boundaries)
- ☐ Look for signs of numerical instabilities in diffusion simulations, such as oscillations or non-physical concentration values, and address any issues that arise.
- ☐ Test sensitivity to small composition changes
- ☐ Compare ternary diffusion behavior to binary limits

Implementing Mobility Data to .tdb File



End-Members for γ Phase

	$\phi_i^{\text{Ni,FCC}}$	$\phi_i^{\text{Fe,FCC}}$	$\phi_i^{\text{Nb,FCC}}$
Ni	$D_0 = 3.44 \text{ cm}^2 \text{ s}^{-1}$ $Q = 290.0 \text{ kJ mol}^{-1}$ [1]	$D_0 = 1.12 \text{ cm}^2 \text{ s}^{-1}$ $Q = 283.5 \text{ kJ mol}^{-1}$ [1]	$D_0 = 3.44 \text{ cm}^2 \text{ s}^{-1}$ $Q = 290.0 \text{ kJ mol}^{-1}$ [1]
Fe	$D_0 = 9.20 \text{ cm}^2 \text{ s}^{-1}$ $Q = 297.7 \text{ kJ mol}^{-1}$ [1]	$D_0 = 0.72 \text{ cm}^2 \text{ s}^{-1}$ $Q = 278.7 \text{ kJ mol}^{-1}$ [1]	$D_0 = 9.20 \text{ cm}^2 \text{ s}^{-1}$ $Q = 297.7 \text{ kJ mol}^{-1}$ [1]
Nb	$D_0 = 0.0104 \text{ cm}^2 \text{ s}^{-1}$ $Q = 202.59 \text{ kJ mol}^{-1}$ [2]	$D_0 = 0.75 \text{ cm}^2 \text{ s}^{-1}$ $Q = 264 \text{ kJ mol}^{-1}$ [3]	$D_0 = 0.0104 \text{ cm}^2 \text{ s}^{-1}$ $Q = 202.59 \text{ kJ mol}^{-1}$ [2]

Species i

CALPHAD Mobility Parameter:

$$\phi_i^p = -Q_i + RT \ln D_0^i$$

The CALPHAD mobility parameter used to model the diffusion of multi-component systems is defined for each species in the phase we want to model diffusion for (γ -Ni which is named FCC_A1 in the .tdb file)

```
$ =====
$ MOBILITY PARAMETERS for FCC_A1 phase
$ =====
$
$ --- Ni mobility in FCC_A1 ---
PARAMETER MQ(FCC_A1&NI,NI:VA;0) 298.15 -290000.0+R*T*LN(3.44E-4); 6000 N !
PARAMETER MQ(FCC_A1&NI,FE:VA;0) 298.15 -283500.0+R*T*LN(1.12E-4); 6000 N !
PARAMETER MQ(FCC_A1&NI,NB:VA;0) 298.15 -290000.0+R*T*LN(3.44E-4); 6000 N !

$ --- Fe mobility in FCC_A1 ---
PARAMETER MQ(FCC_A1&FE,NI:VA;0) 298.15 -297700.0+R*T*LN(9.20E-4); 6000 N !
PARAMETER MQ(FCC_A1&FE,FE:VA;0) 298.15 -278700.0+R*T*LN(7.21E-5); 6000 N !
PARAMETER MQ(FCC_A1&FE,NB:VA;0) 298.15 -297700.0+R*T*LN(9.20E-4); 6000 N !

$ --- Nb mobility in FCC_A1 ---
PARAMETER MQ(FCC_A1&NB,NI:VA;0) 298.15 -202590.0+R*T*LN(1.04E-6); 6000 N !
PARAMETER MQ(FCC_A1&NB,FE:VA;0) 298.15 -264000.0+R*T*LN(7.50E-5); 6000 N !
PARAMETER MQ(FCC_A1&NB,NB:VA;0) 298.15 -202590.0+R*T*LN(1.04E-6); 6000 N !
$
```

Binary Diffusion Profiles to Model



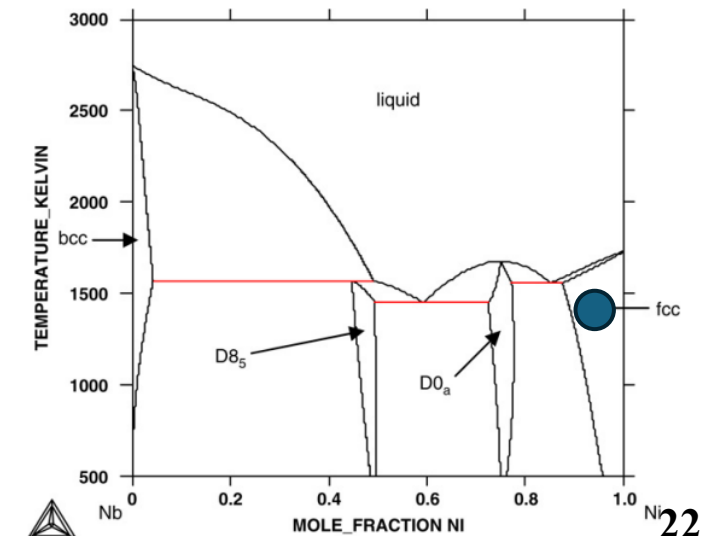
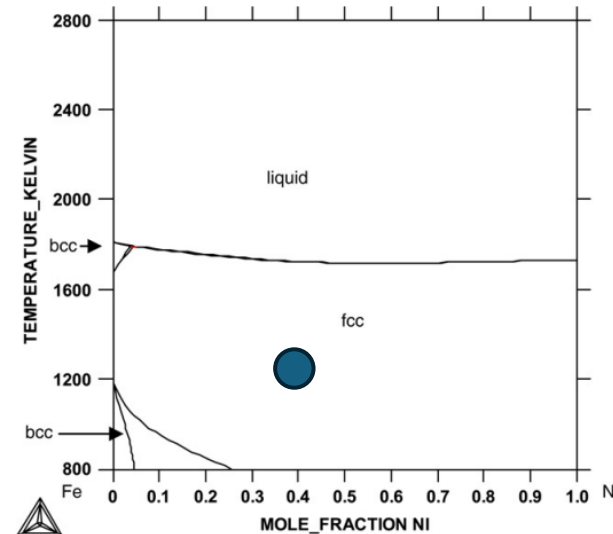
IN718 Limiting Composition [%]

Nickel (plus Cobalt)	50.00-55.00
Chromium.....	17.00-21.00
Iron	Balance*
Niobium (plus Tantalum).....	4.75-5.50
Molybdenum	2.80-3.30
Titanium.....	0.65-1.15
Aluminum	0.20-0.80
Cobalt	1.00 max.
Carbon	0.08 max.
Manganese	0.35 max.
Silicon	0.35 max.
Phosphorus.....	0.015 max.
Sulfur.....	0.015 max.
Boron	0.006 max.
Copper	0.30 max.



Binary couple compositions and temperatures modeled ensured system remained in the FCC_A1 phase because this is the only phase mobility data was defined for in the .tdb.

The Fe-Nb binary couple was not modeled because of the low solubility of Nb in FCC Fe



Binary Diffusion Profiles – Fe-Ni



Setup:

Overall Couple Length: 1 mm

Holding Time: 20 hrs

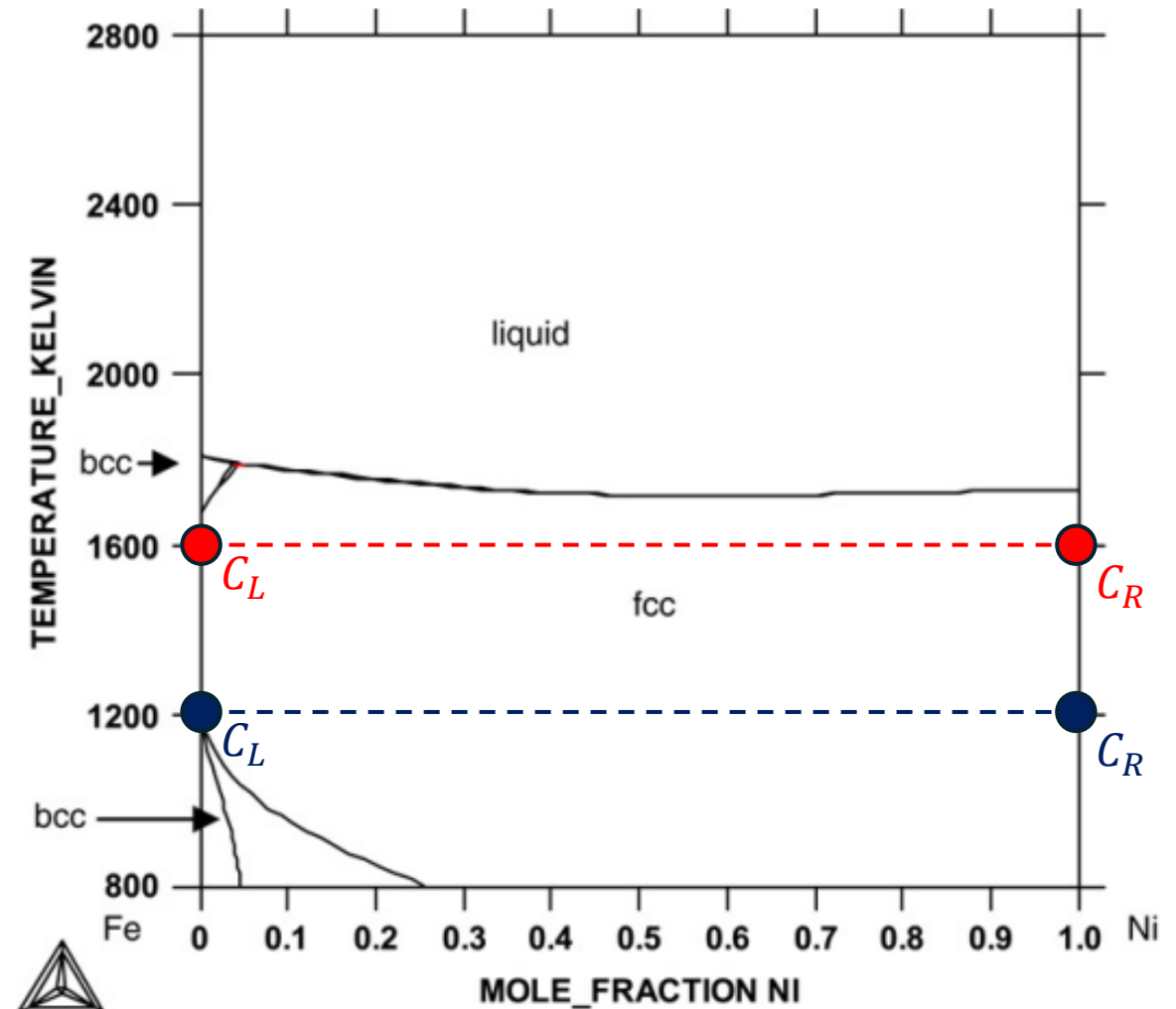
C_L : 100 at % Fe, 0 at % Ni

C_R : 0 at % Fe, 100 at % Ni

of Grids: 200

Run 1 Temperature: 1600 K

Run 2 Temperature: 1200 K

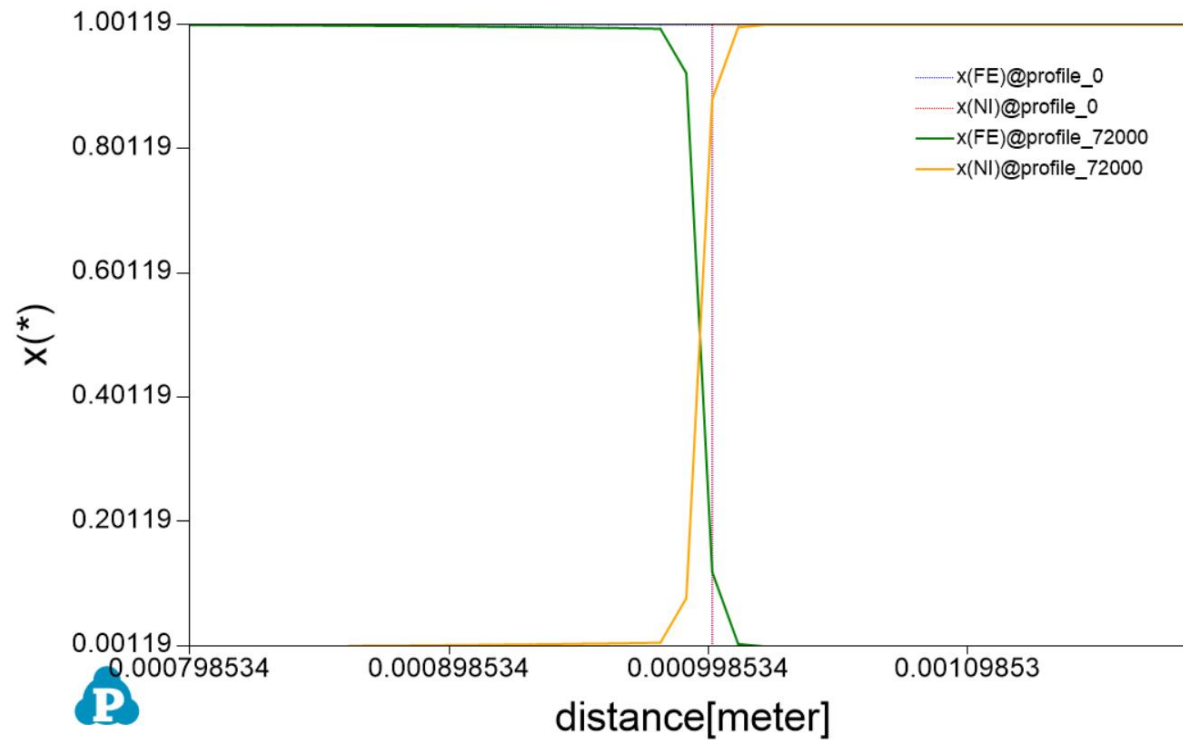


Binary Diffusion Profiles – Fe-Ni

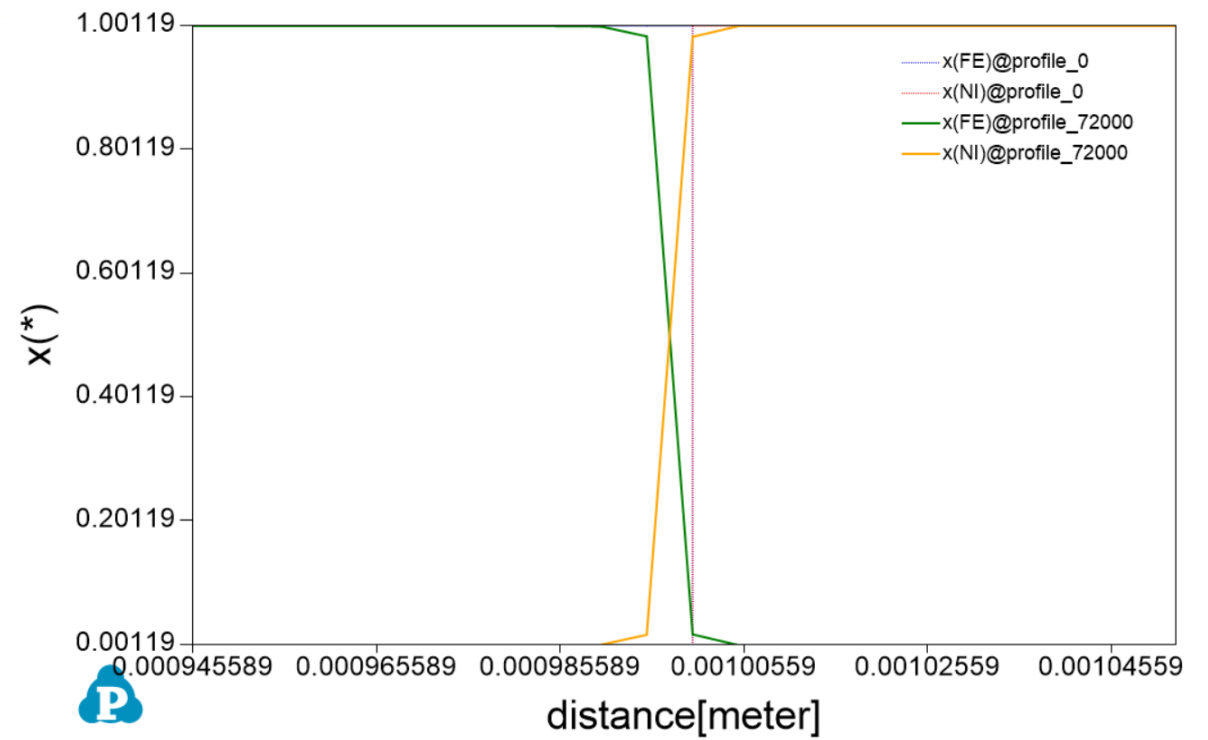


Results:

Run 1



Run 2



Binary Diffusion Profiles –Ni-Nb



Setup:

Overall Couple Length: 1 mm

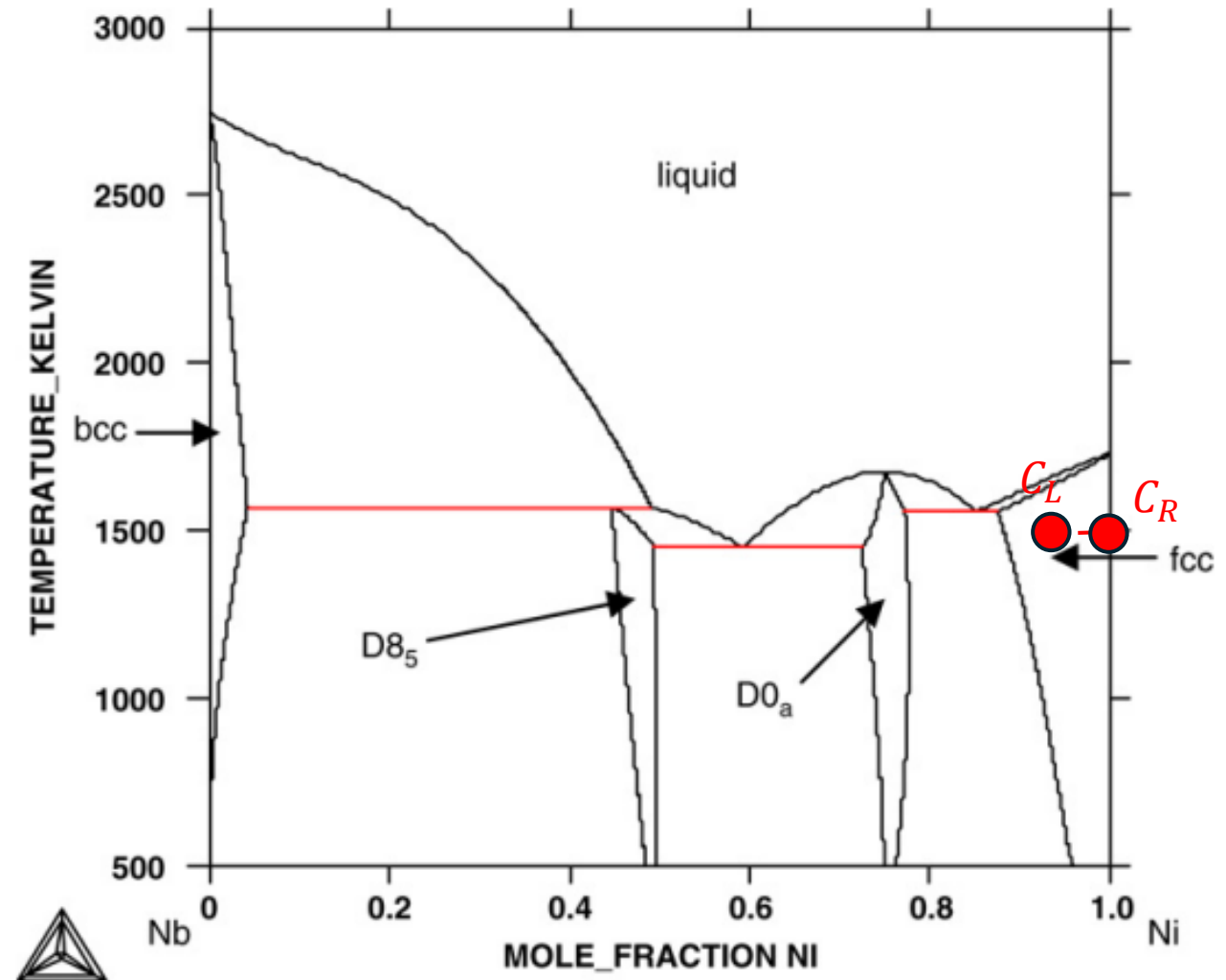
Holding Time: 20 hrs

C_L : 95 at % Ni, 5 at % Nb

C_R : 100 at % Ni, 0 at % Nb

of Grids: 400

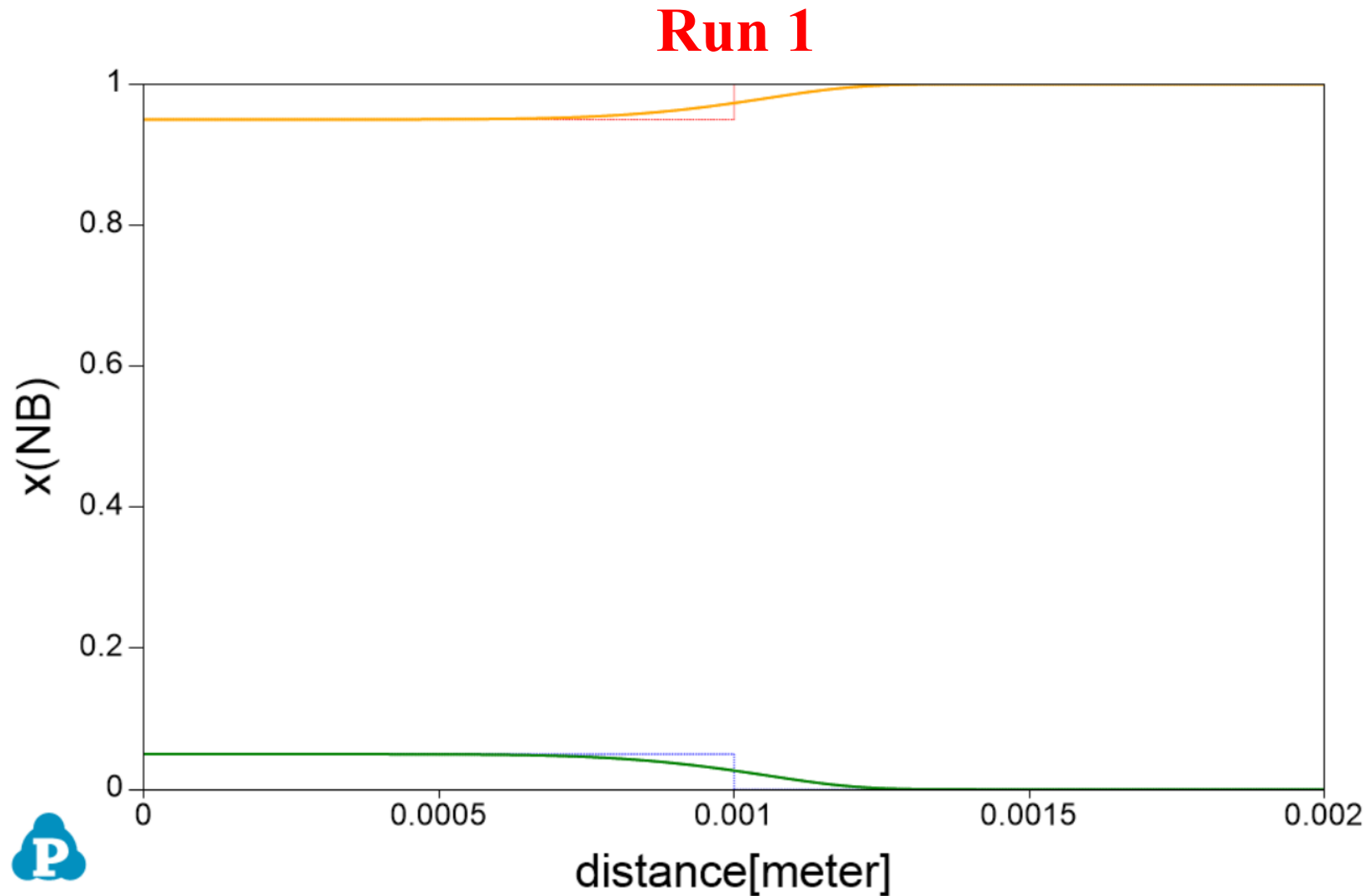
Run 1 Temperature: 1500 K



Binary Diffusion Profiles – Ni-Nb



Results:



Ternary Diffusion Profiles



Setup:

Overall Couple Length: 1 mm

Holding Time: 20 hrs

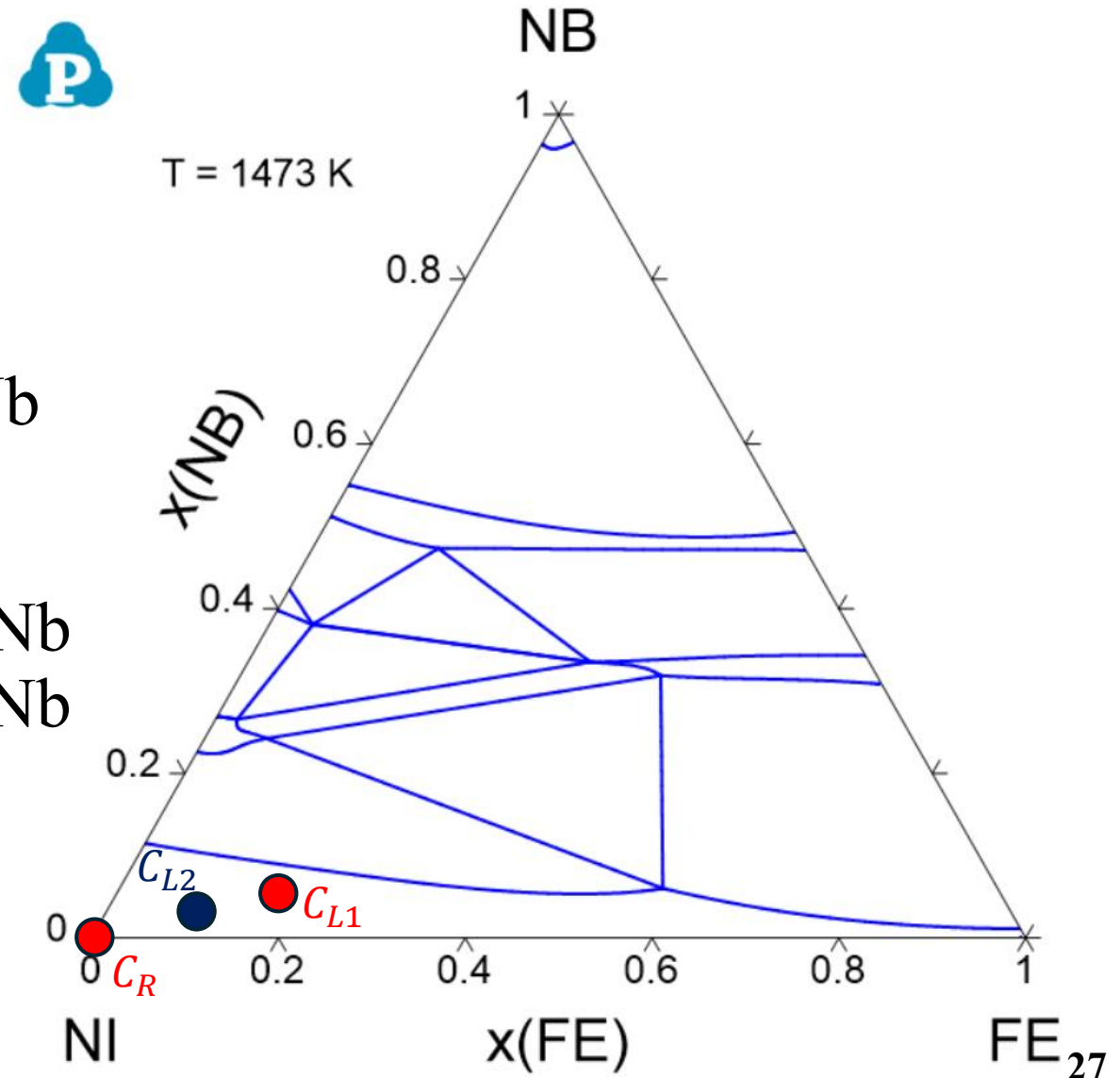
Temperature: 1473 K

C_R : 100 at % Ni, 0 at % Fe, 0 at % Nb

of Grids: 400

C_{L1} : 75 at % Ni, 20 at % Fe, 5 at % Nb

C_{L2} : 83 at % Ni, 10 at % Fe, 7 at % Nb

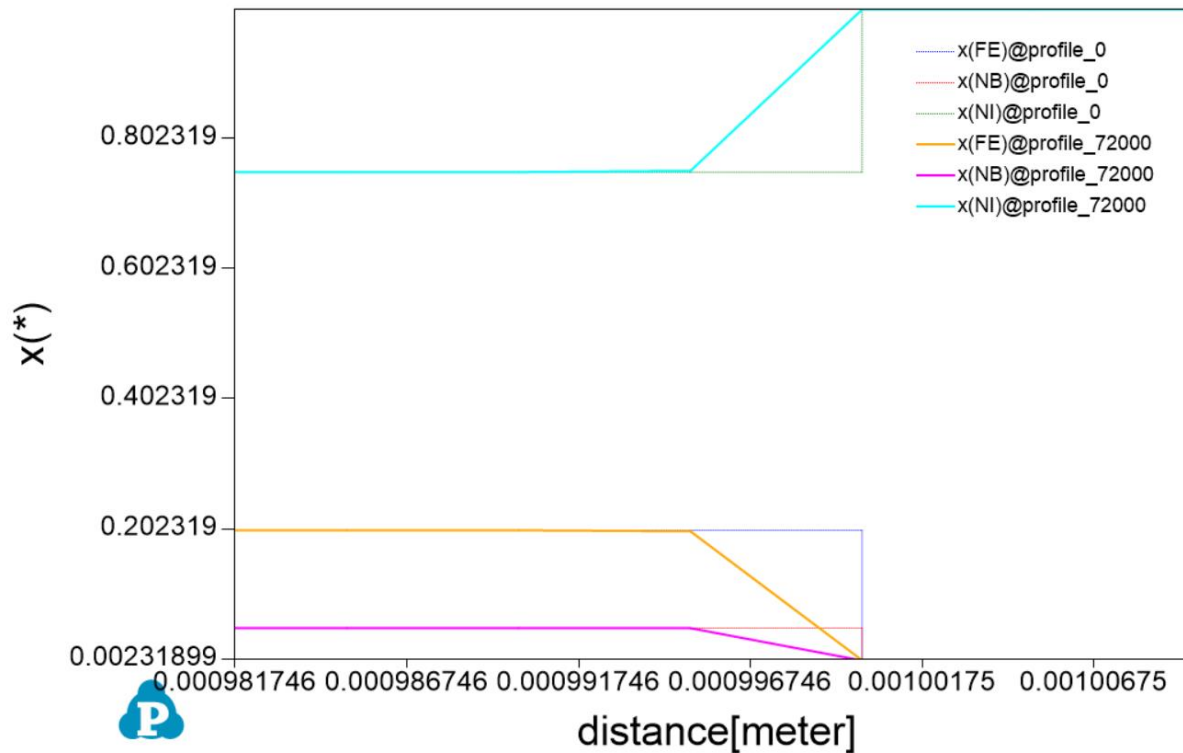


Ternary Diffusion Profiles

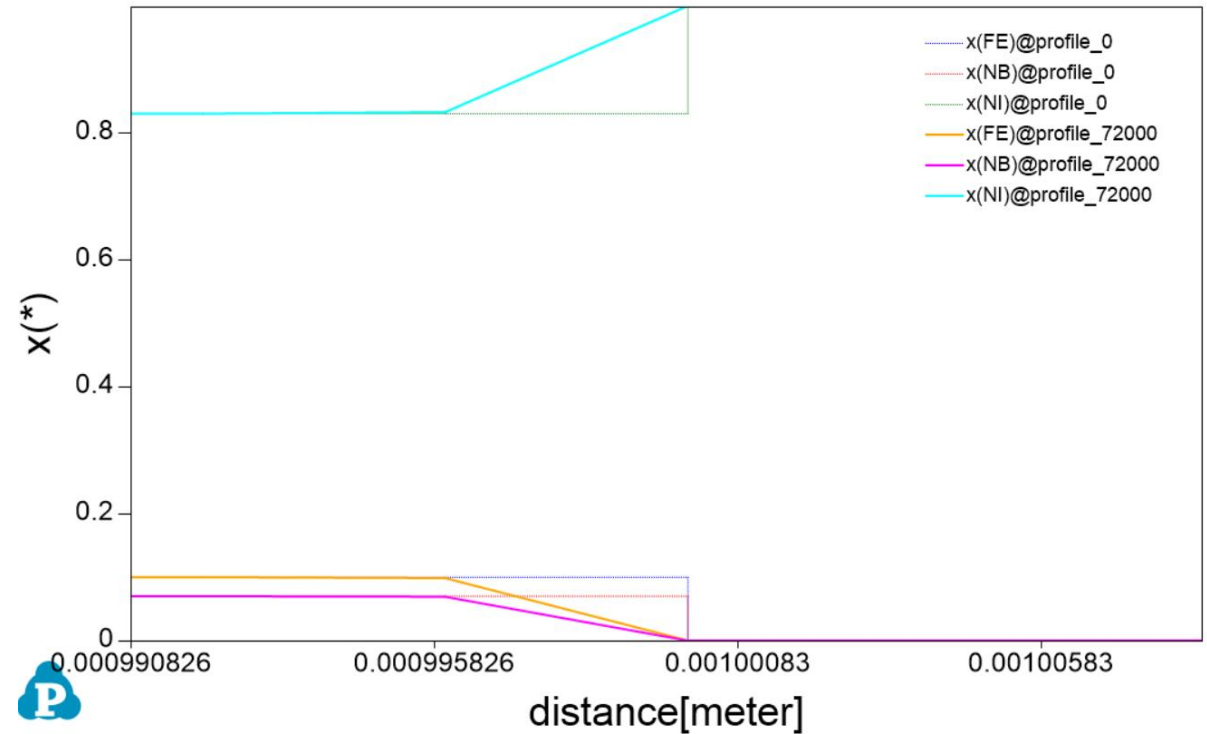


Results:

Run 1



Run 2



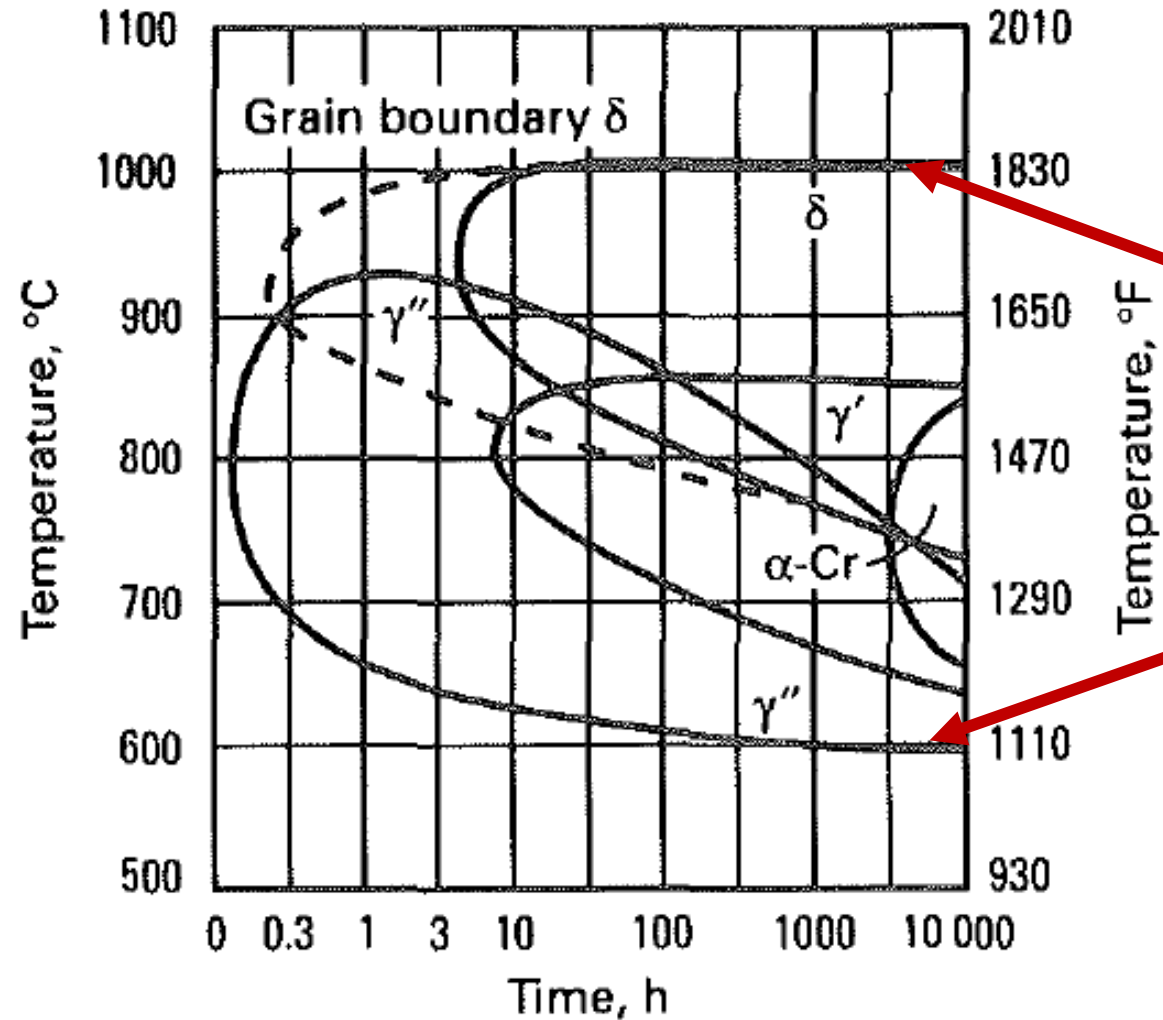
Key Findings



- The physically reasonable binary diffusion profiles and ternary diffusion profiles suggest that the mobility data was successfully implemented for the γ -Ni (FCC_A1) phase in the Fe-Ni-Nb thermodynamic database
- The concentration profiles calculated for the Ni-Nb diffusion couple was the “smoothest” concentration profile because Nb has a higher diffusivity in γ -Ni than Ni or Fe
 - Higher diffusivity improves the spatial resolution of the concentration profile calculated for comparable holding times and temperatures and a fixed number of grids (i.e. a fixed grid spacing)
- Longer holdings times and a smaller grid spacing is required to improve the spatial resolution of the ternary concentration profile, which will be relevant to the TTT calculations

IV. Calculating TTT and CCT Diagrams

TTT diagram I'm trying to re-create...



Chandler H. 1996

Kinetic Mobility Database



- Have made a kinetic mobility database to calculate the precipitation of γ'' and δ from a γ -Ni matrix
- .kdb imports into PanEvolution/PanPrecipitation
- When single-temperature precipitation or TTT calculations are attempted, Pandat fails to calculate any precipitation of γ'' or δ
- Currently trying to debug why the .kdb specifically is causing this issue since the mobility data in the .tdb has proved to work (diffusion profiles)

```
<?xml version="1.0" encoding="iso-8859-1"?>
<PPREC version="1.0">

<Header copyright="User">
  <!-- Kinetic Database of PanPrecipitation for IN718 TTT Simulation -->
  <!-- Matrix: FCC_A1 (gamma) -->
  <!-- Precipitates: D022_NBNi3 (gamma''), D0A_NBNi3 (delta) -->
  <Application name="PanPrecipitation" version="1.0" />
</Header>

<Alloy name="IN718_KWN">

  <MatrixPhase name="FCC_A1">

    <ParameterTable type="kinetic" name="Parameters for gamma">
      <!-- Using Standard Unit (SI) -->
      <Parameter type="Molar_Volume" value="7.1E-6" description="Molar Volume" />
    </ParameterTable >

    <!-- gamma'' (D022_NBNi3): coherent, intragranular, low interfacial energy -->
    <PrecipitatePhase name="D022_NBNi3" model="KWN" morphology="Sphere" nucleation="Modified_Homo" growth="Simplified">
      <ParameterTable type="kinetic" name="Parameters for gamma_double_prime">
        <!-- Using Standard Unit (SI) -->
        <Parameter type="Molar_Volume" value="7.5E-6" description="Molar Volume" />
        <Parameter type="Interfacial_Energy" value="0.025" description="Interfacial Energy" />
        <Parameter type="Atomic_Spacing" value="3.60E-10" description="Atomic Spacing" />
        <Parameter type="Nucleation_Site_Parameter" value="0.01" description="Nucleation Site Parameter" />
        <Parameter type="Driving_Force_Factor" value="1.0" description="Driving Force Factor" />
      </ParameterTable >
    </PrecipitatePhase >

    <!-- delta (D0A_NBNi3): incoherent, grain boundary, higher interfacial energy -->
    <PrecipitatePhase name="D0A_NBNi3" model="KWN" morphology="Sphere" nucleation="Grain_Boundary" growth="Simplified">
      <ParameterTable type="kinetic" name="Parameters for delta">
        <!-- Using Standard Unit (SI) -->
        <Parameter type="Molar_Volume" value="7.5E-6" description="Molar Volume" />
        <Parameter type="Interfacial_Energy" value="0.3" description="Interfacial Energy" />
        <Parameter type="Atomic_Spacing" value="3.60E-10" description="Atomic Spacing" />
        <Parameter type="Nucleation_Site_Parameter" value="0.001" description="Nucleation Site Parameter" />
        <Parameter type="Driving_Force_Factor" value="1.0" description="Driving Force Factor" />
      </ParameterTable >
    </PrecipitatePhase >

  </MatrixPhase >
</Alloy >

</PPREC>
```


Kinetic Mobility Database Testing



- Attempting single-temperature precipitation calculations first to see if by kinetic mobility database can be used to calculation precipitation in general (example below)

This calculation time is highly suspicious, but I've not figured out a root cause yet

Calculation Duration: 0.3 seconds

Calculation Type: precipitation

of databases: 2

tdb: C:\Users\Anirudh Bali\Desktop\Spring 2026\MTEN 6020 - Materials Kinetics\Project Materials\.tdb\Fe-Nb-Ni_baliad_00.TDB

kdb: C:\Users\Anirudh Bali\Downloads\IN718_precipitation_v2.kdb

Composition:

x[FE] = 0.22

x[NB] = 0.03

x[NI] = 0.75

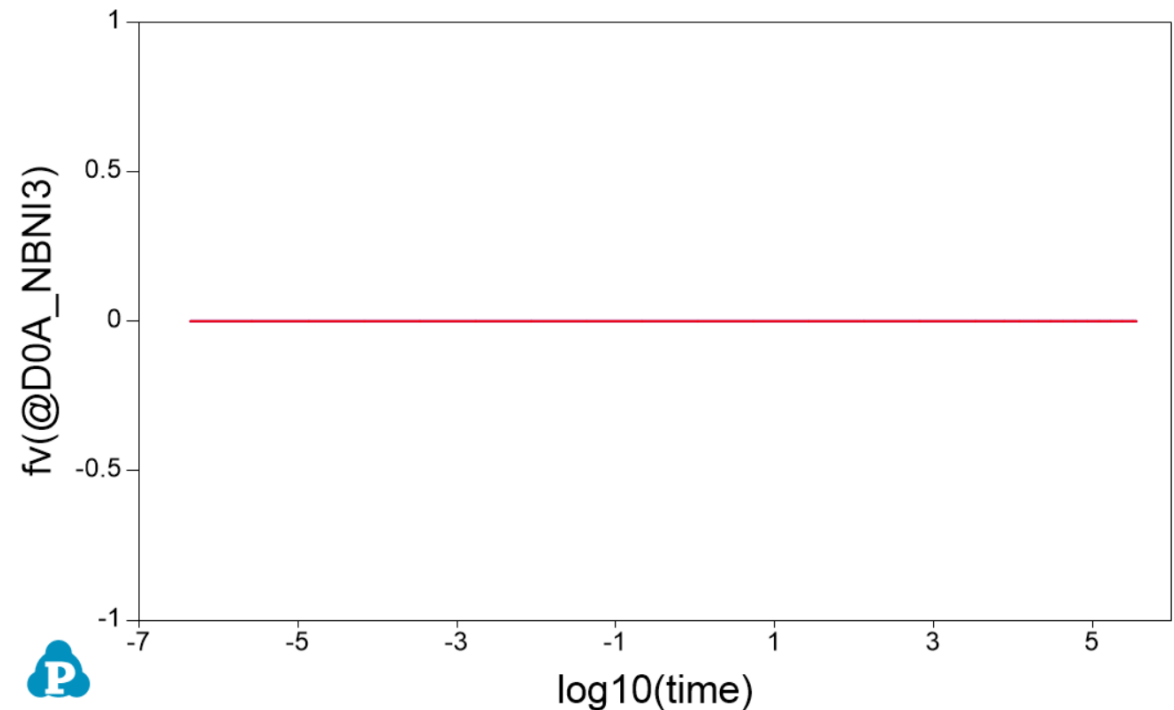
Processing History:

time = 0 [hour]

T = 950 [C]

time = 100 [hour]

T = 950 [C]



Kinetic Mobility Database Testing



What I'm working through to try to correct precipitation calculations:

- **Equilibrium Calculations** – It is possible that the IN718 TTT is misleading me about how long it takes for γ'' or δ take to precipitate from the matrix when the other alloying elements are not considered (see slide 22 for full list of alloying elements featured in IN718)
- Test higher temperatures and longer holds for single-temperature precipitation calculations
- Manually try to find the nose of the TTT curves I'm looking for once I'm calculating a non-zero volume fraction for either target phase

References



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